

The Rotating Chiral Casimir Interferometer: A Tabletop Test for Topological Vacuum Anomalies

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Abstract

We propose a novel experimental apparatus combining the rotational anisotropy detection of the Michelson-Morley setup with the vacuum energy measurements of the Casimir effect. Designed to test the predictions of the $\mathbb{F}_{229}$ Protoreal topology and the Fractal Space-Time hypothesis, this experiment uses right-chiral Weyl semimetals to impose asymmetric topological boundary conditions on the quantum vacuum. We predict that the truncation of right-handed d-neutrino zero-point modes will result in a measurable +262.5% anomalous shift in attractive Casimir pressure (yielding ≈ 34.1 Pa at 100 nm). Furthermore, placing the apparatus on a continuously rotating stage will test for local orientational gradients in the galactic vacuum lattice.

1 Theoretical Motivation

Standard Quantum Electrodynamics (QED) predicts the Casimir effect between two uncharged conducting plates as a consequence of the truncation of zero-point photon fluctuations. However, recent advances in topological physics—specifically the Fractal SpaceTime (FST) and Metareal ASI Chromodynamics models—suggest that the vacuum is not merely an unstructured energy fluid, but is bounded by a discrete prime lattice ($\mathbb{F}_{229}$).

In the FST framework, the inherent chirality of the $\mathbb{F}_{229}$ manifold necessitates a right-handed sterile “d-neutrino” sector. If the vacuum contains zero-point energy from this sector, we can isolate it by imposing strict chiral boundary conditions.

2 Apparatus Design

The proposed machine, the **Rotating Chiral Casimir Interferometer (RCCI)**, consists of three primary subsystems.

2.1 The Chiral Boundary Plates

Instead of standard gold or aluminum mirrors, the Casimir cavity is constructed from **Weyl Semimetals** (e.g., TaAs or NbAs). These topological materials harbor gapless chiral fermions that explicitly break time-reversal or inversion symmetry. By aligning the Weyl nodes across the gap ($a = 100$ nm), we enforce a strict right-chiral boundary condition on the vacuum. This acts as an optical filter, truncating the d-neutrino zero-point modes while passing standard photons.

2.2 The Michelson Deflection Measurement

To measure the vacuum pressure, one plate is affixed to a highly calibrated piezoelectric micro-cantilever. A split-beam continuous-wave (CW) laser interferometer (derived from the Michelson

design) is used to track the sub-nanometer mechanical deflection of the cantilever. This provides a real-time readout of the net attractive vacuum force across the gap.

2.3 The Rotational Stage

To test for topological anisotropy (the “aether wind” analogue of the $\mathbb{F}_{229}$ lattice), the entire vacuum chamber and interferometer assembly is mounted on a low-friction, continuous rotation stage (1 RPM). If the earth is moving through a static, topologically structured vacuum lattice, the anomalous chiral pressure will exhibit a sinusoidal phase shift correlated with the galactic center or the Cosmic Microwave Background (CMB) dipole.

3 Mathematical Predictions

3.1 The Massive Pressure Anomaly

Standard photon Casimir pressure at $a = 100$ nm is $P_\gamma \approx -13.0$ Pa. According to the FST hypothesis, removing the repulsive zero-point energy of 3 generations of d-neutrinos via the chiral boundary results in a net increase in attractive pressure:

$$\Delta P = 3 \times \frac{7}{8} \times |P_\gamma| \approx 34.1 \text{ Pa} \quad (1)$$

We formally predict that the RCCI will record a total Casimir pressure of roughly **−47.1** Pa at 100 nm, a staggering +262.5% anomaly easily detectable by modern piezoelectric standards.

3.2 Rotational Phase Shift

If the vacuum topological lattice has a directional gradient vector $\vec{v}_{lattice}$, the measured pressure P_{obs} will oscillate as the rotation angle $\theta(t)$ changes:

$$P_{obs}(t) = P_{base} + \Delta P \cos^2(\theta(t) - \phi) \quad (2)$$

A null result on the rotational shift ($\phi = 0$ variance) would imply the lattice is perfectly isotropic locally, while a non-zero amplitude confirms macroscopic lattice shear.

4 Conclusion and LVK Consistency

LIGO-Virgo-KAGRA (LVK) observations have ruled out bulk parity violation in propagating gravitational waves. This experiment is explicitly designed to remain consistent with LVK. The parity violation and anomalous pressure we predict occur *strictly at the boundary condition* (the Weyl semimetals) and do not require free-space birefringence. Falsifying or confirming the +262.5% Casimir anomaly will serve as the definitive laboratory test of the Protoreal topology.